

AOC-0 / R-1 Flight operations manual

Suborbital procedures and checklists



About this document

This document describes the procedures to be used during a AOC-0 suborbital mission. This mission requires an R-1 launcher and all known mission variants are described in this document. This document also contains the checklists and actions to be taken under unusual and emergency circumstances.

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Mission types

There are 3 known mission types for the AOC-0 in a suborbital configuration: exo-atmospheric exploration, precision targeted launch and exo-atmospheric tourism. Both exo-atmospheric missions types are similar, with the tourism mission having an unqualified tourist instead of a qualified crewman.

Final assembly

Mount AOC-0 capsule on top of decoupler.

Parachutes should align to R-1 parachutes

Hatch should align with R-1 gas generator exhaust

Hatch should align with R-1 fin

Orient capsule/booster stack with capsule hatch pointing towards desired direction of gravity turn

Usually east

AOC-0 causes booster to tilt towards hatch side

Re-root to Mk1 Command pod

Staging should be set to decouplers and parachutes in stage 0 and engine in stage 1.

Quality control steps

If the decoupler has been separated from the booster, ensure that it is reconnected to the booster and capsule. If the decoupler is connected to the heat shield and the booster is then connected to the capsule, it is possible that the booster ends up connected directly to the heat shield via the second node in the heat shield, making it impossible to separate. This can be seen when the booster causes the shroud to appear around the heat shield despite the decoupler being connected to the heat shield.

Quick turnaround procedure

After a suborbital mission and recovery of booster and capsule.

1. Mate capsule and booster
2. Charge capsule battery
3. Enable decoupler staging
4. Charge propellants on booster
5. If capsule batteries were drained, recharge computer battery.

Consider having an extra capsule ready for use. If the capsule has been serviced prior to mating, the KCT turnaround time is reduced from 1 hour and 20 minutes to 37 minutes.

Suborbital exo AOC-0 launch profile

This profile is essentially a vertical flight up to the limits of the Kerbin atmosphere. Booster is ejected as soon as the craft leaves the atmosphere and the craft and booster are automatically recovered via passive mechanisms.

Flight parameters

Launcher R-1

Max altitude 105Km

Vertical flight

Equipment loadout

None. Remove resources and leave stores are indicated below.

Resource	Amount	Units	Notes
Electrical charge	FULL	EC	
Spare parts	0	Units	Mission too short for repairs
Food	EMPTY	%	LS is not an issue for short flights
Water	EMPTY	%	LS is not an issue for short flights
Oxygen	EMPTY	%	LS is not an issue for short flights
Carbon Dioxide	EMPTY	%	Should not need to be drained
Waste water	EMPTY	%	Should not need to be drained
Waste	EMPTY	%	Should not need to be drained

Pilot should carry a parachute in exploration missions. Tourism missions do not carry parachutes (they actually do, but cannot use them).

Life support is not necessary in suborbital missions because missions are shorter than 15 minutes while kerbals can survive 2 hours without oxygen (breathing the air in the capsule, I suppose). After the 15 minute flight, air is taken from Kerbin's atmosphere, extending endurance. The limit is 18 hours because of lack of water. Recovery should be performed within that time.

Pre-rollout check

- Verify equipment loadout
 - No spare parts
 - No monopropellant
 - Life support EMPTY
 - Waste EMPTY
 - Pilot carries parachute, tourists do not
 - No EVA jetpack
- Verify R-1 propellant levels
 - Fully loaded

- Verify R-1 parachute parameters
 - Pressure 0.6
 - Altitude 500
 - Angle 10
 - Open when safe
- Verify AOC-0 parachute parameters
 - Pressure 0.6
 - Altitude 750
 - Angle 10
 - Open when safe
- Verify electrical system
 - Fully charged, both in Stayputnik and Mk1 pod
 - EC flow set to 0 in Mk1, -1 in Stayputnik, priority goes to Mk1
- Verify that booster decoupler is stageable
- Verify that no Dangit reports indicate damage
- Verify staging
 - Engines on stage 1
 - Parachutes and decoupler on stage 0

In flight checklists and procedures

SYSTEM LIST (CAPSULE)

Pod

Reaction wheel

Lights

Battery

Part storage bin

Food storage

Water tank

O2 tank

CO2 tank

Waste tank

Wastewater tank

Computer

Battery

Parachute (L)

Parachute (R)

Radio (L)

Radio (R)

Thermometer

Barometer

Heat shield

Shield

Ejector

SYSTEM LIST (BOOSTER)

Engine

Fins

LFOX tanks

Parachutes

Decoupler

MINIMUM EQUIPMENT LIST

Capsule

Pod

Wheel

Battery

Parachute (L)

Parachute (R)

Heat shield

Ejector

Radio x 2 (tourism only)

Booster

Engine

Fins

LFOX tanks

Parachutes x1

Decoupler

NORMAL PROCEDURES

Pre flight

Computer hibernate	OFF
Radios	ON
Computer EC	FULL
Pod EC	FULL
LFOX Tanks	FULL
Decoupler	STAGEABLE

DANGER: Launching with an unstageable decoupler requires an attached landing which is a potentially deadly situation

Pod Parachutes	CHECK PARAMS (0.6/750/10)
Rocket Parachutes	CHECK PARAMS (0.6/500/10)
Throttle	MAX
Capsule hatch	Oriented east unless otherwise required

DANGER: Launching with the hatch oriented west from KSC may result in capsule landing in mountains

FMRS is not necessary, capsule and booster remain in close proximity during the flight

Launch

Set KER to orbit + surface + land

Landing pointer shows estimated landing position

Perform pre-flight checklist

Stage to start engine

Rocket will veer towards hatch side

Hatch-side forces increase until shortly before MECO

Landing point can be placed within a 25Km circle around KSC by rotating and tilting rocket

Tilt rocket after MECO for final adjustments

At 70 KM

Stage

Capsule lands automatically upon reentering atmosphere. Booster lands before capsule. Recovery process can start immediately.

Mission specific emergency procedures

These procedures are specific for this type of mission only and may be used instead of the general procedures.

Booster LFOX tank failure (boost phase)

Continue burn until propellants are exhausted.

At 70 Km or at apoapsis (if lower):

Stage

Proceed with mission normally.

Suborbital precision AOC-0 launch profile

This profile sacrifices altitude for additional control. Booster is kept until propellants are exhausted and is used to guide the craft to a specific landing point. The craft and booster are automatically recovered via passive mechanisms after decoupling.

Flight parameters

Launcher R-1

Max altitude Variable

Angled flight in lower/mid atmosphere. May reach high atmosphere.

Equipment loadout

Equipment loadout is identical to exo-atmospheric loadout.

Pre-rollout check

The pre-rollout check is identical to the exo-atmospheric check.

In-flight checklists and procedures

System list, and minimum equipment list are identical to exo-atmospheric checks. No provision is made for tourism flights in this regime.

NORMAL PROCEDURES

Pre flight

Computer hibernate	OFF
Radios	ON
Computer EC	FULL
Pod EC	FULL
LFOX Tanks	FULL
Decoupler	STAGEABLE
Pod Parachutes	CHECK PARAMS (0.6/750/10)
Rocket Parachutes	CHECK PARAMS (0.6/500/10)
Throttle	MAX

Launch

Set KER to orbit + surface + land

Landing pointer shows estimated landing position

Use SCANSat screen or map with estimated landing point to estimate error in reaching target

Perform pre-flight checklist

Stage to start engine

Rotate and tilt rocket to guide landing indicator to desired landing point

Rocket will veer towards hatch side

Immediately after launch, rotate to point hatch to landing point and pitch over towards target

Cut thrust as soon as the estimated landing site is close to the target

Overshoot if survey is at high altitude

Tilt to 30 degrees from vertical for maximum range

In the lower atmosphere, use rocket thrust to compensate for drag

As soon as a proper trajectory is achieved, decouple booster.

Capsule lands automatically close to target. Booster lands before capsule. Recovery process can start immediately.

EMERGENCY PROCEDURES

These procedures should be followed in case of unexpected events or damage to the spacecraft and are common to all suborbital operation modes.

General emergency procedures

Pod battery failure

Engine Thrust	CUT
Staging	ACTIVATE
Return to kerbin	

Pod parachute failure (single)

Engine Thrust	CUT
Staging	ACTIVATE
Return to kerbin	

Booster engine failure

Engine Thrust	CUT
Staging	ACTIVATE
Return to kerbin	

Booster LFOX tank failure (boost phase)

Engine Thrust	CUT
Staging	ACTIVATE
Return to kerbin	

Booster decoupler failure

Staging	ACTIVATE
Decoupler	MANUAL DECOUPLE
Verify parachute state (capsule)	DEPLOYED
Verify parachute state (booster)	DEPLOYED
Pod heat shield	JETTISON

Pod parachute failure (dual)

Engine Thrust	CUT
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Conserve propellants and limit apoapsis as much as possible.

Option 1: Decouple and eject in the lower atmosphere, landing by personal parachute.

Option 2: Follow attached re-entry checklist.

Option 1 should be taken if apoapsis is above 55 KM. Option 2 should be taken if option 1 is undesirable or impossible or if large amounts of propellant are available in the booster's tanks.

Heat shield jettison failure

Verify parachute state (capsule) DEPLOYED

Verify parachute state (booster) DEPLOYED

Decoupler MANUAL DECOUPLE

If decouple unsuccessful, follow attached re-entry checklist.

Attached re-entry

Thrust CUT

Capsule parachutes RECONFIGURE

Pressure 0.01

Altitude 5000

Open when risky

Booster parachutes RECONFIGURE

Pressure 0.01

Altitude 5000

Open when risky

Before apoapsis:

Attitude MAX DRAG (nose to horizon)

After apoapsis:

Above 30 KM

Attitude MAX DRAG

30 to 19 KM

Attitude SURFACE RETROGRADE

At 19 KM altitude, burn to reduce velocity to below 500 m/s.

When parachutes open, burn to keep velocity below 300 m/s for as long as possible.

Prevent nose down attitude at all costs.

Crew parachute re-entry

Remain in capsule until atmospheric re-entry is complete.

Return to kerbin